

VirtualBox Cybersecurity Lab

Phase 1 & Phase 2 — Complete Portfolio Documentation

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Objective: Deploy an isolated lab with Kali Linux and Windows 11 on Oracle VirtualBox, connected via custom NAT Network (10.0.0.0/24), verify connectivity with ping tests, and take recovery snapshots.

Final Lab Summary

Component	Details
Host OS	Windows 11 (Physical Machine)
Hypervisor	Oracle VirtualBox
NAT Network	NatNetwork — 10.0.0.0/24 (DHCP Enabled)
Kali Linux	IP: 10.0.0.3/24 Snapshot: Clean Kali - Network Ready - Phase 1
Windows 11	IP: 10.0.0.4/24 Snapshot: Clean Windows 11 - After Install - Phase 2
Ping Test	Windows (10.0.0.4) → Kali (10.0.0.3) — SUCCESS (0% loss)

PHASE 1 — Kali Linux Setup

1.1 Install 7-Zip

Downloaded and installed 7-Zip from the official website to extract the Kali Linux .7z archive.

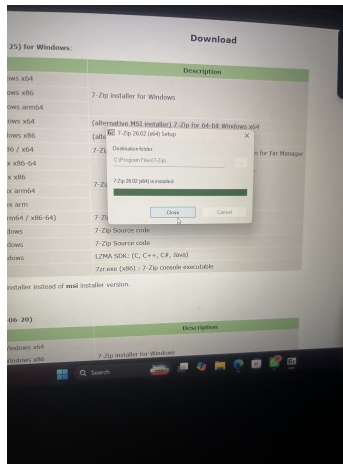


Figure 1: 7-Zip installation

1.2 Create NAT Network (10.0.0.0/24)

Created custom NAT Network named **NatNetwork** with IPv4 prefix 10.0.0.0/24 and DHCP enabled.

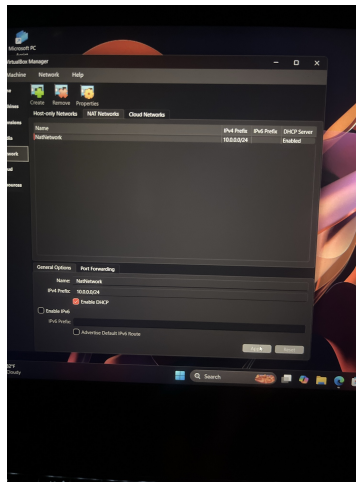


Figure 2: NAT Network configuration (10.0.0.0/24)

1.3 Import Kali Linux Virtual Machine

Downloaded official Kali VirtualBox image, extracted it, and imported the .vbox file into VirtualBox.



Figure 3: Kali Linux successfully imported

1.4 Configure Kali Network Adapter

Set Adapter 1 to **NAT Network** → **NatNetwork**.

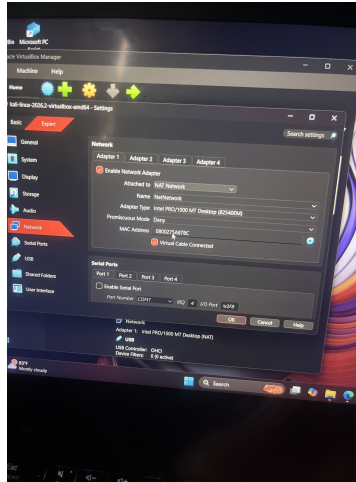


Figure 4: Kali network adapter set to NatNetwork

1.5 Verify IP Address

Logged in with credentials **kali / kali** and ran `ip a`. Obtained IP **10.0.0.3/24**.

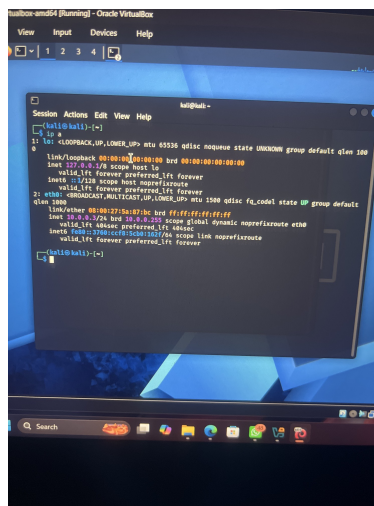


Figure 5: Kali Linux IP address — 10.0.0.3/24

1.6 Take Snapshot

Snapshot name: **Clean Kali - Network Ready - Phase 1**

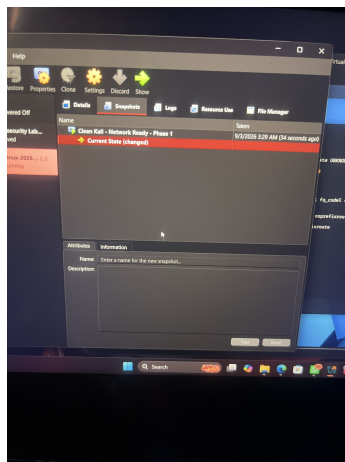


Figure 6: Kali snapshot created

Downloaded official x64 ISO (Win11_25H2_English_x64_v2.iso) from Microsoft.



Created VM with 4 GB RAM, 2 CPUs, UEFI, 80 GB disk, and attached to NatNetwork.



2.3 Install Windows 11

Completed Windows 11 setup (language, edition, partitioning) and reached the desktop.

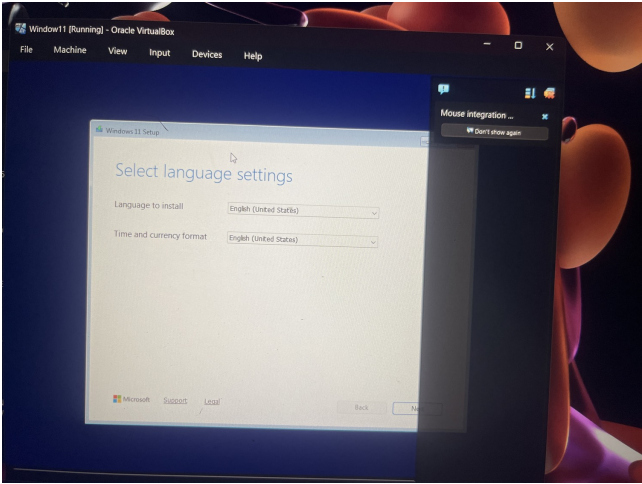


Figure 10: Windows 11 Setup — Language screen

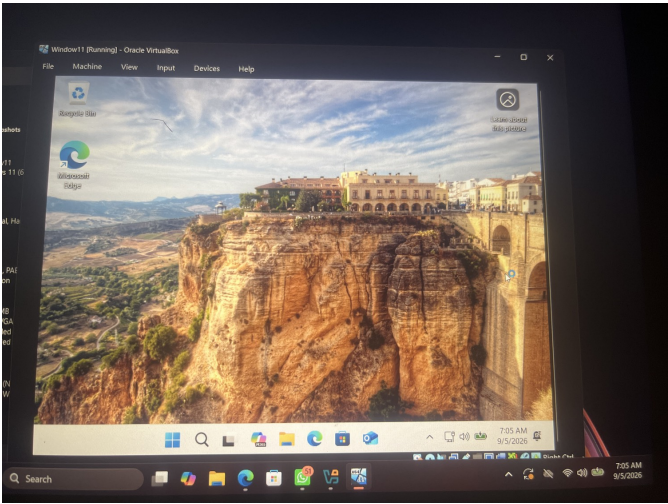


Figure 11: Windows 11 desktop — Installation complete

Connectivity Verification (Ping Tests)

3.1 Both Machines Running

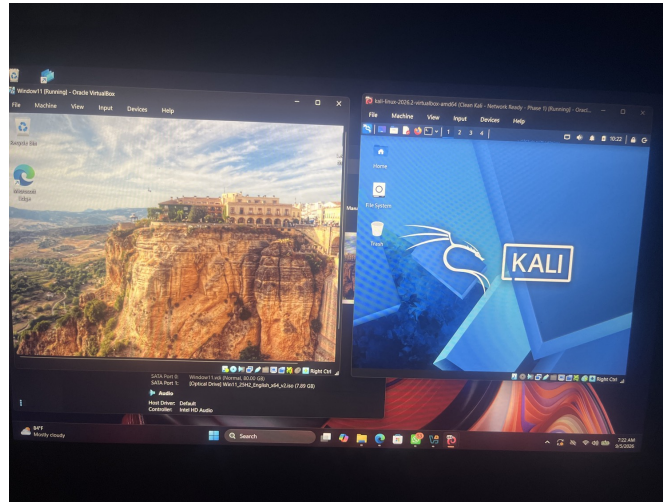


Figure 12: Kali Linux and Windows 11 running side-by-side

3.2 Windows 11 IP Address

Ran `ipconfig` on Windows 11. Assigned IP: **10.0.0.4**

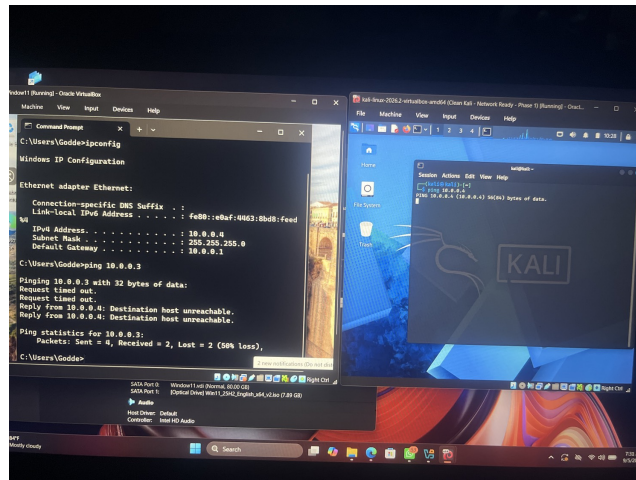


Figure 13: Windows 11 IP configuration — 10.0.0.4

3.3 Successful Ping Test

From Windows 11 (10.0.0.4) pinged Kali Linux (10.0.0.3). Result: **0% packet loss**.

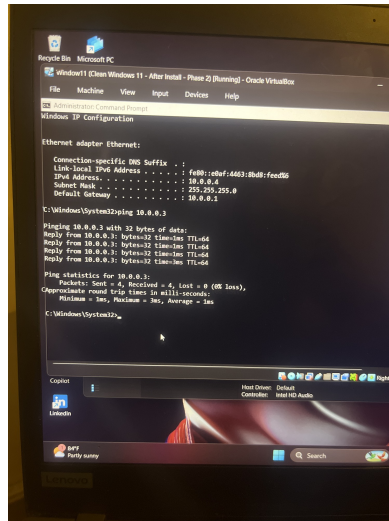


Figure 14: Successful ping from Windows 11 to Kali Linux

PING SUCCESSFUL — Lab network is fully functional

Snapshots

Windows 11 Snapshot

Snapshot name: **Clean Windows 11 - After Install - Phase 2**

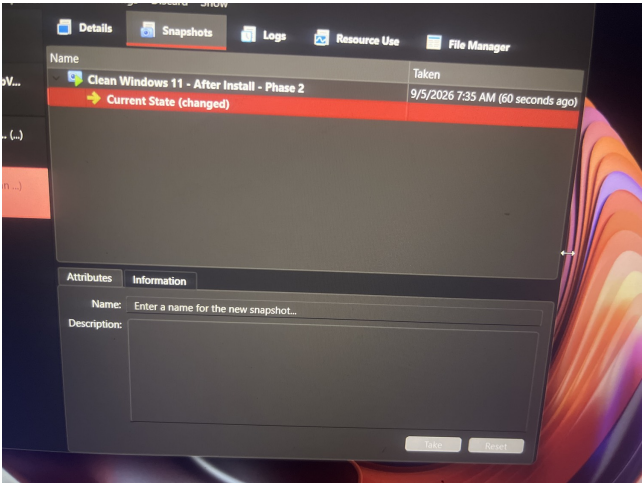
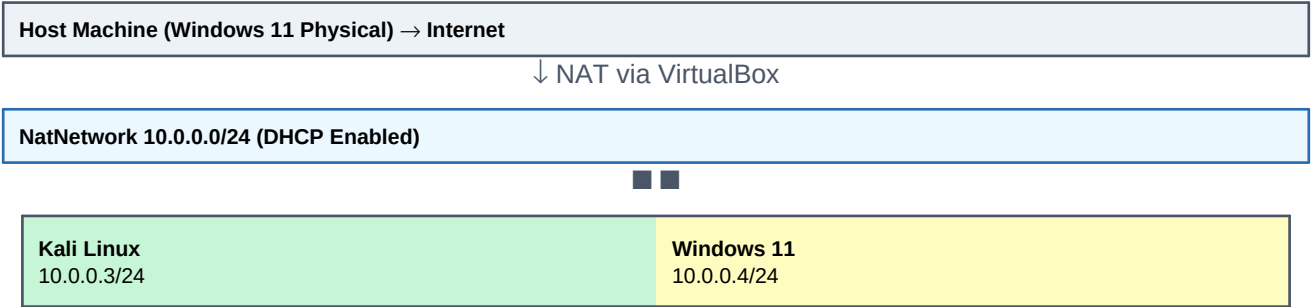


Figure 15: Windows 11 snapshot taken

VM	Snapshot Name	Purpose
Kali Linux	Clean Kali - Network Ready - Phase 1	Clean state after network config (IP 10.0.0.3)
Windows 11	Clean Windows 11 - After Install - Phase 2	Clean state after installation & network

Network Topology



Final Status

Phase 1 — COMPLETED

7-Zip installed • VirtualBox ready • NAT Network (10.0.0.0/24) created • Kali imported • IP 10.0.0.3 configured • Snapshot taken

Phase 2 — COMPLETED

Windows 11 ISO downloaded • VM created • Network attached • Windows 11 installed • IP 10.0.0.4 obtained • Bidirectional connectivity verified (ping success) • Snapshot taken